CARLOS CARDENAS

AI Engineer — Production Systems · Hacker & Builder

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Senior Software Engineer with 10+ years building production software, data platforms and machine learning systems. My focus is the path from research prototype to reliable production service: Python 3.12 backends, optimized Docker images, workloads running on Kubernetes, scheduled pipelines, and SQL over Snowflake, BigQuery and Databricks. I work closely with data science teams to industrialize models, and I care about testing, observability and maintainability more than about adding another framework to the stack.

PRODUCTION ML PLATFORM — WHAT I ACTUALLY DO

- **Python in production.** FastAPI and Flask services on Python 3.10–3.12, typed contracts, pytest suites, structured logging, packaging with uv/poetry.
- **Containers.** Multi-stage Docker builds for applications and ML models; layer-cache strategy, slim runtime images, reproducible builds wired into CI.
- **Kubernetes.** Deployments, Services, health probes, resource limits and HPA on EKS; scheduled retraining, batch scoring and ETL as **CronJobs** with retries, backoff limits, concurrency policy and alerting.
- **Model serving.** Custom containerized inference layer (FastAPI + Docker on Kubernetes): versioned endpoints, request/response logging, latency budgets and drift monitoring. Familiar with the trade-offs against managed serving stacks (BentoML, KServe, Seldon).
- **Data & SQL.** Writing and optimizing analytical SQL on **Snowflake**, BigQuery and Databricks for feature engineering, aggregation and reporting workloads.
- **Automation.** GitHub Actions pipelines: tests, image build, registry push, rolling deploy, infrastructure as code with Terraform.

EXPERIENCE

Tunky
04/2024 – Present

Senior Software Engineer

- Own the backend and ML platform behind seven production products (liti.app, suyay.app, doorhq.app, megol.app, urpix.app, gopues.app, vilif.app), from architecture to on-call.
- Build Python services exposing ML predictions over REST and GraphQL, backed by PostgreSQL and Kafka-based event processing for high-throughput workloads.
- Containerize models and services with multi-stage Docker images and deploy them to Kubernetes (EKS) and AWS Lambda through GitHub Actions.
- Run recurring retraining, batch inference and ETL as Kubernetes CronJobs, designed to be idempotent and safe to re-run after failure.
- Develop ML feature engineering pipelines on Databricks and Apache Spark; query and optimize SQL over Snowflake and BigQuery for feature and analytics layers.
- Build React and Next.js dashboards for monitoring model behaviour and business KPIs.
- Apply a spec-driven development workflow with paired implementer and adversarial-reviewer agents to keep quality high on a small team.

Stack: Python, FastAPI, PostgreSQL, Docker, Kubernetes (EKS), Kafka, Databricks, PySpark, Snowflake, AWS Bedrock, LLMs, GitHub Actions, Terraform, React, Next.js.

RankMi
08/2023 – 04/2024

Engineering Manager

- Led performance work on RankMi's AI platform, cutting latency and infrastructure cost on the critical analytics path.
- Built Genius AI: LLM-based and reinforcement learning components for real-time people analytics, taking them from experiment to served endpoints.
- Deployed serverless ML inference on Vertex AI and AWS Lambda, with containerized model artifacts and automated release pipelines.

Stack: Python, Rust, Databricks, TensorFlow, PyTorch, Pandas, OpenAI, LLMs, AWS, GCP, Vertex AI, Docker.

Leapcure
12/2022 – 08/2023

Senior Software Engineer

- Built integrations between clinical trial systems and data analysis pipelines, with a mobile-first product surface.
- Profiled and restructured API endpoints and data access, substantially reducing response times on the heaviest queries.

Stack: Python, AWS, Data Lake, Apache Spark, Scikit-Learn, ReactJS.

BairesDev
02/2022 – 08/2022

Senior Data Engineer

- Modernized legacy data systems for logistics firms in Australia and Europe: test coverage, containerization and repeatable deployment replaced manual processes.
- Introduced MLOps practices (versioned models, automated retraining, monitoring) to teams that had been running models by hand.

Stack: Python, TensorFlow, Keras, PyTorch, Pandas, Databricks, MLOps, AWS, Azure, NLTK.

Government of Peru — Ministry of Health
04/2021 – 02/2022

Senior Data Engineer

- Ran the monitoring platforms for national health indicators during the COVID-19 emergency, holding data accuracy

within a 1% margin against official sources.
- Delivered call center and job board platforms for Peruvian citizens, operating under public scrutiny and real load.

Stack: Python, PyTorch, TensorFlow, PySpark, ETL, DynamoDB, AWS, Power BI, Blockchain.

Rappi
02/2021 – 03/2021

Senior Software Engineer

- Backend engineer on "Favores y Antojos"; rebuilt the rain forecast module around clear design patterns and refactored a large share of its core codebase.

Stack: Python, PyTorch, Kotlin, Groovy, AWS, Jenkins, Docker, Git.

Globant
09/2020 – 02/2021

DevOps Engineer

- Supported the Smile Direct Club and Disney accounts: endpoint performance tuning, deployment automation and observability improvements.

Stack: Python, Django, TensorFlow, PostgreSQL, Redis, Power BI.

Huawei
01/2015 – 09/2019

Senior Data Software Engineer

- Led delivery for the TUGO and SVA projects at Telefonica, introducing continuous deployment and preventive maintenance that materially improved platform response times.
- Built ETL and reporting pipelines over Hadoop, PySpark and Oracle for large telecom datasets.

Stack: Python, PySpark, Hadoop, TypeScript, React, Pandas, TensorFlow, Oracle, SQL Server, Azure, AWS, NLTK.

Sugar Labs — Google Summer of Code
02/2018 – 08/2018

Mentor Developer

- Mentored two student projects; designed their CI/CD integrations, contribution process and scraping tooling.

Stack: Python, Go, Vue.js, GCP, CI/CD.

EARLIER EXPERIENCE

2012 – 2015	**Tata Consultancy Services Peru** — Software Engineer. Delivery for Rimac Seguros and Strategic Brokers; process redesign and UX research. *Java, Python, Ruby, Oracle, Ext JS, Jenkins.*
2012	**Google Summer of Code — ASCEND4** — Developer. Rebuilt most of the UI of an open-source solver for large linear and non-linear systems. *Python, Fortran, Xcode.*
2012	**Municipalidad de Miraflores** — Systems Engineering Specialist. IT audit of management systems under COBIT best practices. *Python, Django, ACL, IDEA.*
2010 – 2011	**UDEP / Peruvian Navy** — Software Engineer. Real-time navigational aid systems; led Varayoc, a command-and-control system for the fleet. *C++, Qt, Python, Linux.*
2009 – 2010	**Cayetano Heredia University** — Software Engineer. Project management and monitoring for technological architecture. *Python, PHP, Ruby, Java, SQL Server.*
2009	**MCD Consulting S.A.C.** — Software Engineer. Cut processing and transfer times between business systems. *C++, C#, ASP.NET, Oracle.*

EDUCATION

MIT edX MicroMasters — Statistics and Data Science	2021
Harvard edX — Data Science Professional Certificate	2020
CENTRUM, Pontificia Universidad Catolica del Peru — MBA	2012
Universidad Nacional de Ingenieria — Systems Engineering	2008

TECHNICAL SKILLS

Languages: Python (primary, 3.12), SQL, Go, C/C++, JavaScript/TypeScript, Ruby, Scala, R.

Data warehouses & processing: Snowflake, BigQuery, Databricks, Apache Spark / PySpark, Hadoop, Kafka, ETL pipeline design.

Databases: PostgreSQL, MySQL, SQL Server, Oracle, SQLite, DynamoDB, Milvus.

Containers & orchestration: Docker (multi-stage builds, image optimization), Kubernetes (Deployments, CronJobs, HPA, probes), EKS, Helm.

Cloud: AWS (Lambda, EKS, Bedrock, S3), GCP (Vertex AI), Azure, Huawei Cloud, OpenStack.

DevOps: GitHub Actions, Jenkins, GitLab CI, Terraform, Ansible, Puppet, New Relic, Jira, Confluence.

ML & AI: PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, clustering and statistical analysis (K-Means, silhouette scoring), LangChain, LangGraph, RAG, MCP, function calling, structured outputs, agent workflows.

Web: FastAPI, Flask, Django, GraphQL, React, Next.js, Node.js.

Languages spoken: Spanish (native), English (professional working proficiency).